

Research Progress Check Point 2

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

1. Research Progress Overview

During this checkpoint period, I continued the development of my predictive-maintenance research across two complementary tasks: machine-failure classification using AI4I 2020 and remaining-useful-life prediction using NASA C-MAPSS FD001.

The work progressed beyond conventional predictive-model benchmarking towards deployment-oriented model selection. Predictive-maintenance research increasingly considers challenges beyond predictive accuracy, including robustness, deployment conditions and practical model-selection considerations [1], [2], [3], [4].

The completed experiments examined predictive quality, controlled sensor corruption, probability reliability where applicable, computational efficiency, ranking stability, and downstream decision risk. These experiments generated quantitative evidence showing that deployment-oriented criteria can lead to model rankings and recommendations that differ from conventional performance-only evaluation.

An important methodological development was the implementation of Reliability-Aware Deployment Rank Aggregation (RDRA). The purpose of RDRA is to investigate whether the empirical stability of the evidence supporting different deployment dimensions should influence how their criterion-specific rankings are aggregated.

A significant research-design impediment was also found throughout weekly monitoring. Even while there has been a lot of experimental work done, the main methodological contribution needed more explanation. Specifically, the project had to explain why consensus ranking is necessary, what defines acceptable ranking behaviour, why RDRA could be useful in addition to Equal Borda, and how ranking robustness and predictive robustness are related.

Therefore, the project was refined around one central methodological problem:

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking when deployment criteria provide conflicting and differently stable empirical evidence?

The revised structure now treats RDRA as the main methodological contribution, corruption analysis as ranking-robustness validation, normalisation analysis as methodological sensitivity validation, and Pareto/regret analysis as downstream decision interpretation.

2. Weekly Research Progress Diary

Week	Meeting date	Main progress	Main evidence	Impediment / feedback	Action
Week 3	24/07/2026	Deployment-oriented AI4I and C-MAPSS evaluation	Figures 1-2	Multiple criteria produced conflicting model preferences	Investigate consensus ranking
Week 4	31/07/2026	RDRA development and initial ranking comparison	Figures 3-4	Need to justify reliability-aware aggregation against a simple baseline	Retain Equal Borda and design ranking-stability validation
Week 5	07/08/2026	RDRA versus Equal Borda and criterion-subset validation	Figures 5-6	Need to define what makes a ranking “better” and avoid equating stability with superiority	Define stability + responsiveness and extend validation to data-quality uncertainty
Intra-trimester break	14/08/2026	Criterion sensitivity, DRS, normalisation, decision-risk analysis and research-story refinement	Figures 7 onward	Need to connect robustness and downstream analysis to the central RDRA story	Consolidate research around one methodological contribution

3. Week 3 - Deployment-Oriented Experimental Development

3.1 Progress and Achievements

During this week, I consolidated the experimental framework used to evaluate predictive-maintenance model configurations beyond conventional clean-data predictive performance.

For AI4I 2020, Random Forest and XGBoost configurations were evaluated under different imbalance-handling and calibration strategies. The outputs included conventional predictive measures together with deployment-relevant evidence such as latency, throughput, model size, corruption robustness and operational cost-related measures. Alternative imbalance-handling and probability-calibration configurations were evaluated because class imbalance and probability reliability can materially affect machine-learning evaluation and decision-making [6], [7].

For C-MAPSS FD001, the candidate set included Ridge, Random Forest, XGBoost, LSTM and GRU. Computational characteristics such as training time, model size, latency and throughput were evaluated in addition to predictive performance.

For AI4I 2020, the experiments addressed binary machine-failure classification using the AI4I 2020 Predictive Maintenance Dataset [5] and included alternative imbalance-handling and probability-calibration configurations.

For C-MAPSS FD001, the experiments addressed remaining-useful-life prediction using classical machine-learning and recurrent deep-learning alternatives. C-MAPSS is a widely used simulated turbofan-engine degradation benchmark for prognostics and RUL research [8], [9].

One important result was that the conventional and deployment-oriented evaluations did not necessarily recommend the same model. In the C-MAPSS experiment, Random Forest was the conventional winner whereas XGBoost became the deployment-oriented baseline winner.

This established the practical research problem:

When deployment-relevant criteria are introduced, different criteria may favour different configurations, so conventional predictive performance alone may not provide a defensible deployment recommendation.

ID	Model				Strategy		Single_Sample_Latency_P95_ms		Throughput_Samples_Per_Second				
	Variant	Model	Strategy	Latency	Throughput	Latency	Throughput	Latency	Throughput	Latency	Throughput	Latency	Throughput
14	XGBoost none platt	XGBoost	none	14	9.553560	86636.431731							
12	XGBoost none isotonic	XGBoost	none	12	10.068794	71469.094500							
13	XGBoost none none	XGBoost	none	13	8.841185	92179.201717							
9	XGBoost class_weight isotonic	XGBoost	class_weight	9	7.891631	88743.361811							
9	XGBoost class_weight none	XGBoost	class_weight	9	8.477350	89049.854354							
17	XGBoost smote platt	XGBoost	smote	17	9.172242	86174.309261							
4	Random Forest none none	Random Forest	none	4	58.211381	25529.416275							
5	Random Forest none platt	Random Forest	none	5	51.522103	26341.555281							
11	XGBoost class_weight platt	XGBoost	class_weight	11	15.043964	85483.382139							
3	Random Forest none isotonic	Random Forest	none	3	51.829886	26653.882762							
											Mean_Cost_Inflation_AUC	Worst_Condition_Retention_Mean	
Calibration Accuracy Precision Recall F1 ROC_AUC PR_AUC											1.688342	0.414212	
14	platt	0.972080	0.592297	0.812529	0.677265	0.975720	0.810519	14	0.839560	0.770768	1.698084	0.412436	
12	isotonic	0.972087	0.586338	0.826471	0.676655	0.969558	0.765946	12	0.839613	0.770388	1.673778	0.414324	
13	none	0.971980	0.592073	0.812560	0.676022	0.975720	0.810519	13	0.838991	0.771500	1.629070	0.439716	
9	isotonic	0.969333	0.563892	0.812745	0.652447	0.967954	0.738093	9	1.063312	0.779660	1.627311	0.440095	
10	none	0.969267	0.563751	0.812745	0.653949	0.974755	0.785892	10	1.062688	0.779242	1.521292	0.436591	
17	platt	0.970533	0.574800	0.827333	0.652080	0.973235	0.796440	17	1.164276	0.765308	1.653159	0.380716	
4	none	0.969600	0.569033	0.807843	0.651984	0.970135	0.772168	4	2.570640	0.772448	1.638716	0.441089	
5	platt	0.969567	0.559927	0.808824	0.651914	0.970199	0.772168	5	2.571183	0.771630	1.623485	0.382504	
11	platt	0.968533	0.557168	0.814706	0.640438	0.974755	0.785892	11	1.063259	0.780461			
3	isotonic	0.968800	0.553484	0.812745	0.648116	0.964488	0.724874	3	2.573283	0.773893			
											1.639891		
Latency_Per_Sample_Median_ms Single_Sample_Latency_Median_ms											Composite_Corruption_Robustness_Std	Composite_PR_AUC_Robustness_Mean	
14	...	0.012239	6.448300	14	0.025913	0.719640							
12	...	0.015075	6.402519	12	0.024433	0.489529							
13	...	0.010927	6.352952	13	0.021722	0.719640							
9	...	0.011508	5.851255	9	0.029314	0.714547							
10	...	0.011659	6.060307	10	0.029558	0.741449							
17	...	0.012082	6.553517	17	0.017686	0.727297							
4	...	0.039682	50.180797	4	0.018979	0.734629							

3.3 Impediment Identified

The increasing number of deployment dimensions created a model-selection problem. Predictive quality, robustness, reliability, efficiency and operational considerations could favour different candidates, but the experiments did not by themselves explain how these conflicting rankings should be combined.

3.4 Action Following the Meeting

The next stage therefore investigated rank aggregation as a mechanism for creating a consensus ranking from conflicting deployment evidence.

Equal treatment of the criterion rankings was retained as a baseline rather than assuming that a new method would automatically be superior.

4. Week 4 - Development of Reliability-Aware Rank Aggregation (RDRA)

4.1 Progress and Achievements

During this stage, I developed and implemented Reliability-Aware Deployment Rank Aggregation (RDRA).

The methodological motivation was that the evidence supporting different deployment dimensions may not exhibit the same empirical stability. Equal aggregation treats the contributing criterion rankings equally, whereas RDRA investigates whether evidence stability can be used to influence their aggregation.

The five deployment dimensions were:

- predictive quality
- corruption robustness
- probability reliability
- computational efficiency
- operational impact.

Rank aggregation provides a mechanism for combining multiple individual rankings into a consensus ranking, with Borda count representing a well-established positional aggregation approach [10], [11].

Rank-aggregation literature explicitly identifies Borda as a classic consensus/positional scoring method.

An important methodological distinction is that the RDRA weight represents confidence in the empirical evidence supporting a criterion ranking, rather than the subjective importance of that deployment criterion.

For example, a criterion can be operationally important while its empirical ranking is comparatively unstable. RDRA addresses the reliability of the supporting evidence, whereas stakeholder preferences are treated separately in the later decision-analysis stage.

4.2 Evidence of Achievement

Evidence 3 - Empirical Stability and RDRA Weight Construction

	Dimension	Mean_Kendall_Tau	Stability_Weight_Raw	N_Repetitions	Stability_Weight
0	Predictive	0.681661	0.681661	20	0.215212
1	Robustness	0.451988	0.451988	20	0.142700
2	Reliability	0.793464	0.793464	20	0.250510
3	Efficiency	0.832026	0.832026	20	0.262685
4	Operational	0.390093	0.408257	20	0.128894

{'Predictive': 0.2152118673045314, 'Robustness': 0.1427002539459913, 'Reliability': 0.2505098109764967, 'Efficiency': 0.26268450524965437, 'Operational': 0.12889356252332607}

Figure 3. Empirical stability estimates and normalised reliability weights used by RDRA across the five deployment dimensions.

The output shows that the five deployment dimensions did not exhibit identical empirical stability.

The estimated normalised RDRA weights were approximately:

Deployment Dimension	Mean Kendall Stability	RDRA Stability Weight
Predictive	0.682	0.215
Robustness	0.452	0.143
Reliability	0.793	0.251
Efficiency	0.832	0.263

Among the evaluated dimensions, Efficiency and Reliability received the largest empirical stability weights, whereas Operational and Robustness received lower weights.

These values do not imply that efficiency is inherently more important than robustness. They indicate that, under the repeated experimental evidence used in this implementation, the corresponding rankings showed different levels of empirical stability.

The results show that the evidence supporting the deployment dimensions was not equally stable. This provides an empirical basis for RDRA weighting rather than arbitrary weighting.

Evidence 4 - Comparison of Aggregated Candidate Rankings

	Variant	Existing_Composite_Score	Existing_Composite_Rank	Equal_Borda_Score	Equal_Borda_Rank	RDRA_Score	RDRA_Rank
14	XGBoost none platt	0.811233	1	0.788235	2	0.836318	1
13	XGBoost none none	0.808227	2	0.788235	1	0.832149	2
12	XGBoost none isotonic	0.791357	3	0.752941	3	0.798244	3
9	XGBoost class_weight isotonic	0.725130	5	0.705882	4	0.713996	4
11	XGBoost class_weight platt	0.744130	4	0.658824	5	0.651132	5
10	XGBoost class_weight none	0.684466	6	0.600000	6	0.583771	6
17	XGBoost smote platt	0.619150	7	0.494118	8	0.559870	7
15	XGBoost smote isotonic	0.588033	8	0.458824	9	0.521118	8
3	Random Forest none isotonic	0.556127	11	0.505882	7	0.497487	9
5	Random Forest none platt	0.564393	10	0.458824	9	0.464931	10
4	Random Forest none none	0.548208	12	0.411765	12	0.397666	11
0	Random Forest class_weight isotonic	0.539454	13	0.435294	11	0.397015	12
16	XGBoost smote none	0.465335	15	0.364706	14	0.381622	13
2	Random Forest class_weight platt	0.565439	9	0.388235	13	0.347607	14
1	Random Forest class_weight none	0.496431	14	0.364706	14	0.317587	15
8	Random Forest smote platt	0.405408	16	0.294118	17	0.267042	16
6	Random Forest smote isotonic	0.386370	17	0.294118	16	0.246996	17
7	Random Forest smote none	0.258919	18	0.235294	18	0.185452	18

Figure 4. Comparison of candidate rankings produced by the existing composite method, Equal Borda and RDRA.

The ranking-comparison output demonstrates that the aggregation method can affect the ordering of candidate configurations.

For example, the leading candidates remain dominated by XGBoost configurations, but differences occur in their exact ordering between the existing composite method, Equal Borda and RDRA.

Under RDRA, XGBoost | none | Platt was ranked first, followed by XGBoost | none | none and XGBoost | none | isotonic.

The purpose of this output is not to claim that a different ranking is automatically better. Instead, it establishes that different aggregation assumptions can produce different model-selection conclusions, creating the need for systematic validation of which aggregation method exhibits the most appropriate behaviour for the research problem.

4.3 Week 5 - RDRA Versus Equal Borda

4.3.1 Purpose of the Comparison

A major milestone during this checkpoint was the empirical comparison between Reliability-Aware Deployment Rank Aggregation (RDRA) and simpler aggregation baselines.

The principal baseline is Equal Borda, which provides a transparent and interpretable method for combining criterion-specific rankings. Equal Borda treats each contributing ranking equally and

therefore represents an appropriate reference point for determining whether incorporating empirical evidence stability provides meaningful additional value.

Equal Borda was selected as a baseline because Borda count is an established and interpretable positional rank-aggregation approach for constructing consensus rankings [10], [11]. Criterion-subset perturbation was used to systematically change the deployment-criterion set and recompute the resulting rankings.

The purpose of this comparison is not to prove that RDRA is universally superior to Borda. Instead, the evaluation asks a narrower and more defensible research question:

Under changes to the deployment-criterion set, does reliability-aware aggregation produce a consensus ranking that exhibits stronger overall ranking consistency than equal aggregation?

This distinction is important because a different ranking is not automatically a better ranking. Therefore, the comparison focuses on several complementary properties of ranking behaviour rather than only the identity of the top-ranked candidate.

4.3.2 Progress and Achievements

Equal Borda was retained as the principal baseline because it is simple, transparent and treats the contributing criterion rankings equally.

The evaluation therefore asked:

Does reliability-aware aggregation provide stronger ranking consistency than equal aggregation when the deployment-criterion set changes?

Criterion-subset perturbation produced 26 ranking protocols across different combinations of the five deployment dimensions.

4.4 Evidence of Achievement

Ranking stability was evaluated using complementary measures including Kendall rank agreement, Spearman rank agreement, top-k preservation, rank displacement and winner survival. Kendall's rank correlation provides a classical measure of agreement between two rankings [12].

Evidence 4 - full ranking-stability comparison

	Method	Mean_Kendall	Mean_Spearman	Mean_Top3_Overlap	Mean_Absolute_Rank_Displacement	Winner_Survival_Rate	N_Protocols
0	Equal_Borda	0.689717	0.806980	0.730769	2.181624	0.230769	26
1	Existing_Composite	0.638512	0.751687	0.756410	2.354701	0.384615	26
2	RDRA	0.742081	0.862428	0.820513	1.782051	0.384615	26

Figure 5. Ranking-stability comparison of Equal Borda, the existing composite method and RDRA under criterion-subset perturbation.

The results show that RDRA achieved:

- the highest mean Kendall agreement
- the highest mean Spearman agreement
- the highest mean top-3 overlap
- the lowest mean absolute rank displacement.

RDRA also achieved a higher winner-survival rate than Equal Borda.

However, the winner-survival rate was equal to that of the existing composite method at approximately 0.385.

This distinction is important because the evidence does not support the claim that RDRA is superior to every baseline on every measure.

Instead, the strongest supported conclusion is that RDRA demonstrates the best overall full-ranking stability profile across the complementary measures used in this experiment.

Evidence 5 - Multi-Level Stability by Number of Dimensions

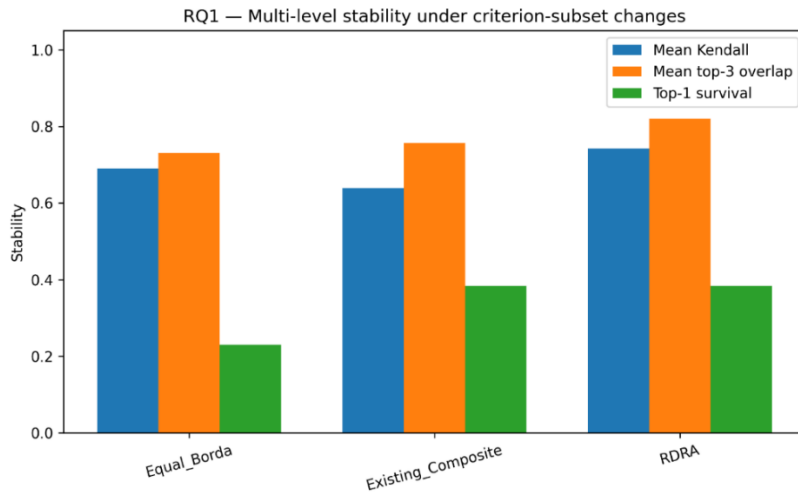


Figure 6. Multi-level ranking stability of Equal Borda, the existing composite method and RDRA as the number of included deployment dimensions changes.

The results show that ranking consistency generally improves as more deployment dimensions are retained.

For the two-dimension subsets, RDRA achieved a mean Kendall agreement of approximately 0.620, compared with approximately 0.542 for Equal Borda and 0.471 for the existing composite.

For three-dimension subsets, RDRA increased to approximately 0.773, compared with 0.726 for Equal Borda and 0.672 for the existing composite.

For four-dimension subsets, RDRA achieved approximately 0.875, again exceeding Equal Borda at approximately 0.851 and the existing composite at approximately 0.835.

When all five dimensions were retained, all methods naturally reproduced their own full reference ranking, resulting in perfect agreement.

This pattern suggests that RDRA's advantage is most relevant when the available deployment evidence is incomplete or reduced rather than when the entire original criterion set is preserved.

4.5 Impediment Identified

There was a significant methodological issue:

More stable does not automatically mean better.

A ranking that never changes could appear highly robust while failing to respond to meaningful changes in deployment evidence. This led to a refined desired property:

A defensible ranking should resist unnecessary instability while remaining responsive to meaningful changes in deployment evidence.

4.6 Action Following the Meeting

The next stage therefore moved from criterion uncertainty to data-quality uncertainty by testing whether RDRA ranking behaviour changed appropriately under controlled sensor corruption.

5. Intra-trimester break - Extended Validation and Research Refinement

5.1 Purpose of the Analysis

After comparing RDRA and Borda generally, I performed a leave-one-dimension-out sensitivity analysis.

The purpose was to determine whether any one deployment characteristic is disproportionately important to the overall RDRA agreement.

This analysis provides a more interpretable view of the aggregation process because it examines how the ranking responds when one source of deployment evidence is removed.

The experiment therefore addresses the question:

Which deployment dimensions are most influential in maintaining the RDRA consensus and the identity of the top-ranked configuration?

This is important because a method can appear stable overall while still relying heavily on one dimension.

5.2 Leave-One-Dimension-Out Procedure

The full RDRA ranking was first established using all five deployment dimensions.

Each dimension was then removed individually:

- Predictive
- Robustness
- Reliability
- Efficiency
- Operational

The RDRA ranking was recomputed after each removal.

The resulting subset ranking was then compared with the full RDRA consensus using:

- Kendall agreement
- Spearman agreement
- mean absolute rank displacement
- top-3 overlap
- winner survival
- subset winner
- and full-reference winner.

This allows both whole-ranking stability and exact recommendation sensitivity to be examined.

5.3 Evidence of Achievement

Evidence 6 - Leave-One-Dimension-Out Results

Removed_Dimension	Kendall_Agreement	Spearman_Agreement	Mean_Absolute_Rank_Displacement	Top3_Overlap	Winner_Survived	Subset_Winner	Full_Winner
2	Reliability	0.790850	0.931889	1.333333	1.0	True	XGBoost none platt
0	Predictive	0.830065	0.942208	1.444444	1.0	False	XGBoost none none
3	Efficiency	0.856209	0.962848	1.111111	1.0	False	XGBoost none none
4	Operational	0.934641	0.985552	0.555556	1.0	False	XGBoost none none
1	Robustness	0.960784	0.991744	0.333333	1.0	True	XGBoost none platt

Figure 7. Leave-one-dimension-out sensitivity analysis showing the effect of removing individual deployment dimensions on agreement with the full RDRA consensus.

The full RDRA winner was XGBoost | none | Platt.

When Reliability or Robustness was removed, this winner remained unchanged.

When Predictive, Efficiency, or Operational was removed, the top-ranked configuration changed to XGBoost | none | none.

5.4 Dimension Influence

The leave-one-dimension-out results also provide insight into the relative structural influence of the deployment dimensions.

Removing Robustness produced the smallest overall ranking disruption, with:

- Kendall agreement ≈ 0.961
- mean rank displacement ≈ 0.333
- the original winner retained.

Removing Operational also resulted in high whole-ranking agreement, but the winner changed.

In contrast, removing Reliability produced the lowest Kendall agreement among the five conditions, at approximately 0.791, although the original winner still survived.

This demonstrates that influence on the overall ranking structure and influence on the top recommendation is not necessarily the same.

A deployment dimension can substantially affect the ordering of lower-ranked candidates without changing the winner, while another dimension can cause a winner change even though the rest of the ranking remains highly similar.

This reinforces the need to evaluate ranking behaviour at multiple levels.

5.5 Weekly Progress - Sensor Corruption and Ranking Robustness

5.6 Progress and Achievements

The robustness analysis was extended from predictive degradation to the stability of the deployment ranking itself.

Controlled corruption and perturbation testing provides a systematic way of evaluating model behaviour when input quality deteriorates [13]. In this research, this principle is extended from predictive-performance degradation to the stability of the resulting deployment ranking.

Two distinct questions were separated:

Predictive robustness: How much does model performance deteriorate?

Ranking robustness: Do those changes materially alter the deployment ordering or recommendation?

The analysis is as follows:

Clean deployment evidence -> Clean RDRA ranking -> Sensor corruption -> Updated evidence -> Recomputed RDRA ranking -> Comparison with clean consensus

This directly addresses the connection between ranking and robustness that later became important in supervision.

5.7 Evidence of Achievement

Evidence 7 - AI4I Deployment-Rank Stability by Severity

	Condition	Level	Kendall_DRS	Spearman_DRS	Scenario_Winner	Clean_Winner	Top1_Survival	Clean_Winner_In_Scenario_Top3	Top3_Set_Overlap
0	Gaussian Noise	0.05	0.921569	0.979360	XGBoost none platt	XGBoost none platt	True	True	1.0
1	Gaussian Noise	0.10	0.986928	0.997936	XGBoost none none	XGBoost none platt	False	True	1.0
2	Gaussian Noise	0.20	0.856209	0.952528	XGBoost none none	XGBoost none platt	False	True	1.0
3	Gaussian Noise	0.30	0.777778	0.900929	XGBoost none platt	XGBoost none platt	True	True	1.0
4	Missing Values MCAR	0.05	0.816993	0.923633	XGBoost none platt	XGBoost none platt	True	True	1.0

Figure 8. AI4I deployment-ranking stability under controlled sensor corruption across severity levels, including Kendall agreement, Spearman agreement, top-1 survival and top-3 preservation.

The AI4I results show that ranking agreement changes as corruption severity increases. For example, under Gaussian noise, the clean winner XGBoost | none | Platt is preserved at severity 0.05, changes to XGBoost | none | none at severity 0.10 and 0.20, and returns to the clean winner at severity 0.30.

Despite these winner changes, the clean winner remains within the top three across the displayed Gaussian-noise conditions, with top-3 overlap equal to 1.0.

This demonstrates that winner stability and broader candidate-set stability can behave differently.

Evidence 8 - AI4I Condition-Level DRS Summary

	Condition	DRS_AUC_Kendall	Top1_Survival_AUC	Top3_Survival_AUC	Worst_Kendall_DRS	Top1_Survival_Rate	Top3_Survival_Rate
0	Gaussian Noise	0.898693	0.416667	1.0	0.777778	0.50	1.0
1	Missing Values MCAR	0.764706	0.416667	1.0	0.686275	0.50	1.0
2	Outliers	0.638562	0.400000	1.0	0.542484	0.50	1.0
3	Static Sensor Offset	0.898693	0.416667	1.0	0.830065	0.25	1.0

Figure 9. AI4I condition-level deployment-rank stability summary across Gaussian noise, missing values, outliers and static sensor offset.

Based on the result, the condition-level summary shows that ranking robustness differs across corruption families.

Gaussian noise and static sensor offset produced higher Kendall-based DRS AUC values of approximately 0.899, while missing values and outliers produced lower values of approximately 0.765 and 0.639, respectively.

The worst Kendall agreement was also lowest for outliers, at approximately 0.542.

In contrast, the top-3 survival rate remained 1.0 across all four corruption conditions, indicating that the leading candidate set was substantially more stable than the exact top-ranked recommendation.

This provides quantitative evidence that robustness cannot be characterised adequately using only winner survival or predictive-retention metrics.

Evidence 9 - C-MAPSS Deployment-Rank Stability

	Condition	Level	Kendall_DRS	Spearman_DRS	Scenario_Winner	Clean_Winner	Top1_Survival
0	Gaussian_Noise	0.05	0.200000	0.000000	Ridge	XGBoost	False
1	Gaussian_Noise	0.10	0.000000	-0.100000	Ridge	XGBoost	False
2	Gaussian_Noise	0.20	0.316228	0.205196	Ridge	XGBoost	False
3	Gaussian_Noise	0.30	0.800000	0.900000	XGBoost	XGBoost	True
4	Gradual_Offset_Drift	0.05	0.200000	0.200000	XGBoost	XGBoost	True

Figure 10. C-MAPSS FD001 deployment-ranking stability under controlled sensor-data corruption, showing changes in rank agreement and top-1 winner survival.

The C-MAPSS results demonstrate a different ranking-stability pattern from AI4I. For the displayed Gaussian-noise conditions, the clean winner is XGBoost. At severity levels 0.05, 0.10 and 0.20, Ridge becomes the scenario winner, while XGBoost returns as the winner at severity 0.30.

Kendall agreement also changes substantially across these severity levels, increasing from approximately 0.20 at severity 0.05 to 0.80 at severity 0.30. This demonstrates that deployment-ranking robustness is both task-dependent and corruption-dependent.

Evidence 10 - AI4I Rank-Stability Visualisation

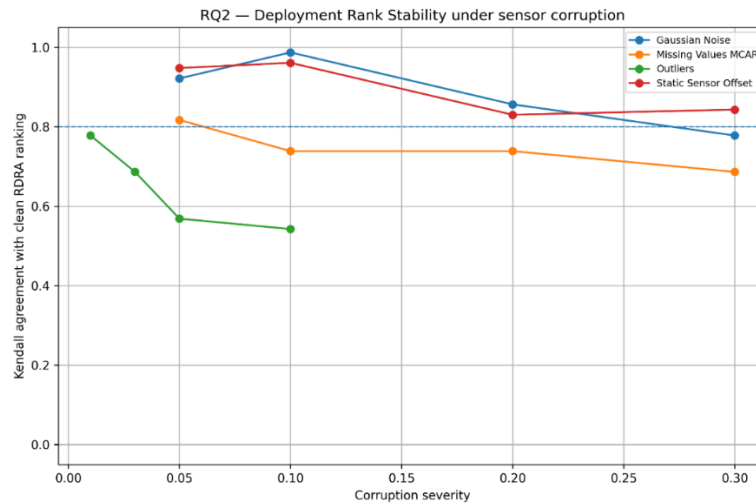


Figure 11. AI4I deployment-ranking stability under controlled sensor-data corruption across increasing severity levels.

The AI4I visualisation shows that deployment-ranking stability depends on both corruption type and severity. Gaussian noise and static sensor offset generally maintain relatively high Kendall agreement with the clean RDRA consensus, although agreement declines at some higher severity levels. Missing-values corruption produces moderate agreement, while outliers cause the largest deterioration in ranking agreement among the evaluated corruption families. The dashed reference line at 0.8 provides a useful visual threshold for comparing stronger and weaker ranking agreement. Overall, the figure demonstrates that deployment-ranking robustness is condition-dependent rather than uniformly preserved across all types of sensor degradation.

Evidence 11 - C-MAPSS Deployment-Rank Stability

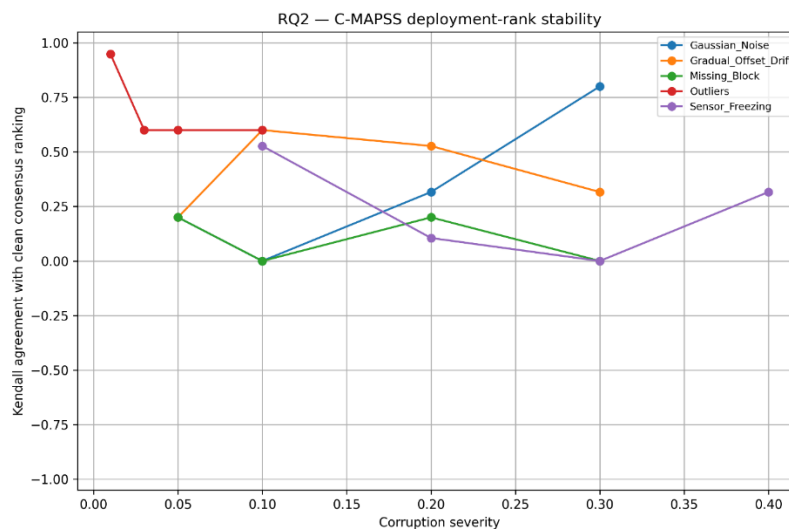


Figure 12. C-MAPSS FD001 deployment-ranking stability under controlled sensor-data corruption across increasing severity levels.

The C-MAPSS visualisation shows substantially greater variation in ranking stability across corruption conditions than AI4I. Gaussian noise produces low agreement at lower severity levels before increasing to approximately 0.8 at severity 0.30. Gradual offset drift shows moderate agreement that declines at higher severity, while missing-block and sensor-freezing conditions produce relatively low agreement across several tested levels. Outlier corruption begins with high agreement at the lowest severity but falls to around 0.6 as severity increases. These results indicate that the C-MAPSS consensus ranking is more sensitive to the type and severity of temporal sensor corruption, reinforcing that ranking robustness is task- and condition-dependent.

6. Methodological Sensitivity Validation

6.1 Motivation and Validation Objective

To determine whether the observed Deployment Rank Stability (DRS) results were overly dependent on the normalisation process used to create the deployment evidence, an additional methodological validation was carried out after the sensor-corruption and deployment-ranking analysis in Section 5.

Because the deployment-ranking system integrates evidence assessed on many scales, this validation was necessary. Without a suitable transformation or normalisation process, it is impossible to directly evaluate predictive quality, corruption robustness, probability reliability, computational efficiency, and operational penalty.

Hence, modifications to normalisation may modify the deployment evidence's relative contribution and, consequently, the RDRA ranking.

6.2 Validation Strategy

The validation compares deployment-ranking behaviour under alternative normalisation treatments.

The key distinction is between normalisation based on the clean-data reference and normalisation that can change according to the corrupted scenario.

Conceptually, the validation asks whether the ranking-stability conclusion remains similar when the normalisation assumption is changed:

Clean deployment evidence -> Establish reference normalisation -> Apply sensor corruption -> Recalculate deployment evidence -> Apply alternative normalisation protocol -> Recompute RDRA ranking -> Compare DRS results

Because the reference scale is set before the introduction of corruption, the clean-data-based method is very helpful. This makes it easier to distinguish between changes brought about by individually rescaling the evidence under each corruption scenario and changes in the underlying deployment evidence.

Therefore, rather than being a separate study contribution, the methodological sensitivity analysis serves as a check on protocol sensitivity.

6.3 Evidence of Achievement

Evidence 12 - AI4I Normalisation-Sensitivity Analysis

	Condition	Adaptive_Mean_Kendall	Fixed_Mean_Kendall	Adaptive_Top1_Survival	Fixed_Top1_Survival	Winner_Change_Rate_From_Normalisation
0	Gaussian Noise	0.885621	0.836601	0.50	0.25	0.25
1	Missing Values MCAR	0.745098	0.797386	0.50	0.25	0.50
2	Outliers	0.643791	0.745098	0.50	0.25	1.00
3	Static Sensor Offset	0.895425	0.862745	0.25	0.25	0.00

Figure 13. AI4I methodological sensitivity analysis comparing deployment-rank stability under alternative normalisation treatments.

The AI4I analysis examines whether the previously observed ranking-stability patterns remain broadly consistent when the normalisation procedure is changed.

The purpose of this comparison is not to require identical Kendall agreement values under every protocol. Instead, it evaluates whether the principal conclusions regarding the relative stability of the deployment ranking are strongly dependent on one scaling assumption.

Where similar patterns are observed across alternative treatments, this provides stronger evidence that the DRS conclusions reflect the underlying corruption response rather than being produced solely by the normalisation procedure.

Evidence 13 - C-MAPSS Normalisation Comparison

	Condition	Level	Kendall_DRS	Kendall_DRS_Fixed	Scenario_Winner	Scenario_Winner_Fixed	Winner_Changed_By_Normalisation
0	Gaussian_Noise	0.05	0.200000	0.200000	Ridge	Ridge	False
1	Gaussian_Noise	0.10	0.000000	0.000000	Ridge	Ridge	False
2	Gaussian_Noise	0.20	0.316228	0.316228	Ridge	Ridge	False
3	Gaussian_Noise	0.30	0.800000	0.800000	XGBoost	XGBoost	False
4	Gradual_Offset_Drift	0.05	0.200000	0.000000	XGBoost	XGBoost	False
5	Gradual_Offset_Drift	0.10	0.600000	0.600000	XGBoost	XGBoost	False
6	Gradual_Offset_Drift	0.20	0.527046	0.527046	LSTM	LSTM	False
7	Gradual_Offset_Drift	0.30	0.316228	0.316228	LSTM	LSTM	False
8	Missing_Block	0.05	0.200000	0.200000	XGBoost	XGBoost	False
9	Missing_Block	0.10	0.000000	-0.105409	Ridge	Ridge	False
10	Missing_Block	0.20	0.200000	0.200000	Ridge	Ridge	False
11	Missing_Block	0.30	0.000000	0.000000	Ridge	Ridge	False
12	Outliers	0.01	0.948683	0.948683	XGBoost	XGBoost	False
13	Outliers	0.03	0.600000	0.527046	LSTM	LSTM	False
14	Outliers	0.05	0.600000	0.400000	LSTM	LSTM	False
15	Outliers	0.10	0.600000	0.400000	LSTM	LSTM	False
16	Sensor_Freezing	0.10	0.527046	0.527046	XGBoost	XGBoost	False
17	Sensor_Freezing	0.20	0.105409	0.105409	XGBoost	XGBoost	False
18	Sensor_Freezing	0.30	0.000000	0.000000	XGBoost	XGBoost	False
19	Sensor_Freezing	0.40	0.316228	0.316228	XGBoost	XGBoost	False

Figure 14. Comparison of C-MAPSS deployment-rank stability under alternative normalisation protocols.

This analysis is particularly important because the C-MAPSS experiments produced greater ranking variation than AI4I.

The comparison determines whether the low or variable Kendall agreement observed under conditions such as Gaussian noise, missing blocks, gradual offset drift, outliers and sensor freezing persists when the normalisation treatment is changed.

If substantial instability remains after controlling the normalisation protocol, the result provides stronger evidence that the behaviour reflects genuine sensitivity of the deployment ranking to corrupted sequential sensor evidence rather than simply an artefact of rescaling.

6.4 Impediment Identified

Furthermore, a significant difficulty in understanding deployment-ranking stability was brought to light by the sensitivity analysis. Alternative normalisation techniques do not have to provide the same rankings. Various normalisation decisions reflect varying presumptions on the appropriate scale for deployment evidence. Thus, total invariance to normalisation should not be used to characterise methodological robustness.

Instead, the analysis needs to differentiate between two types of variation: minor quantitative variation, in which specific agreement values change but the overall research conclusion stays the

same, and substantive protocol dependence, in which altering the normalisation technique significantly alters which candidates are deemed stable or desirable.

6.5 Action Following the Meeting

The next stage is to consolidate the evidence from the ranking and robustness experiments into the downstream deployment-decision analysis.

At this point, the research has established three connected layers of evidence:

Criterion uncertainty: RDRA evaluates whether the consensus ranking remains defensible when the criterion set changes.

Data-quality uncertainty: DRS evaluates whether the deployment ranking remains stable when sensor evidence is corrupted.

Protocol uncertainty: normalisation-sensitivity analysis evaluates whether the observed ranking behaviour depends excessively on an important methodological assumption.

The subsequent analysis therefore examines how this validated deployment evidence can support final candidate selection when stakeholder preferences and decision risk are uncertain.

This leads to Section 7 - Downstream Decision-Risk Analysis.

7. Downstream Decision-Risk Analysis

7.1 Progress and Achievements

Following the development and validation of the deployment-ranking methodology, the next stage investigated how the evaluated deployment evidence can support a final model-selection decision when decision-maker preferences are uncertain.

Multi-objective and multi-criteria decision methods provide mechanisms for examining trade-offs when candidate alternatives cannot be evaluated through a single objective [14].

Therefore, this stage addresses a different source of uncertainty from the previous analyses.

In RQ1, the uncertainty concerns the evaluation criteria and their empirical reliability.

In RQ2, the uncertainty concerns sensor-data quality and the resulting stability of the deployment ranking.

The downstream analysis asks a different question: If multiple configurations remain defensible, how sensitive is the final choice to different assumptions about the importance of predictive quality, corruption robustness, efficiency and operational penalty?

To answer this, the analysis returns to the validated deployment objectives, not the RDRA score, to avoid double-counting. It then applies:

Validated evidence -> Pareto screening -> Preference_scenarios -> Candidate_regret -> Mean/tail/worst-case risk -> Final risk-aware recommendation

This produces a preference-robust deployment decision, showing how each candidate behaves across plausible weighting assumptions and downside-risk conditions.

7.2 Evidence of Achievement

Evidence 15 - Regret-Risk Summary

	Variant	Mean_Regret	P95_Regret	CVaR95_Regret	Worst_Case_Regret	Zero_Regret_Frequency	Minimax_Rank	CVaR95_Rank	Mean_Regret_Rank
3	XGBoost class_weight platt	0.096088	0.287290	0.446101	0.887321	0.0000	1	1	2
2	XGBoost class_weight isotonic	0.033974	0.278828	0.455535	0.933845	0.8576	2	2	1
0	Random Forest class_weight isotonic	0.447315	0.873045	0.916135	0.991296	0.0740	3	3	3
1	Random Forest class_weight platt	0.538289	0.969837	0.985266	0.999690	0.0684	4	4	4

Figure 15. Decision-risk comparison of candidate deployment configurations using mean regret, CVaR95 and worst-case regret.

This analysis compares candidate configurations across different levels of decision risk rather than relying only on average utility.

Mean regret identifies candidates that perform consistently well across typical preference scenarios, whereas CVaR95 evaluates performance in the adverse tail of the regret distribution. Worst-case regret provides the most conservative perspective by identifying the maximum decision loss observed for each candidate.

Considering these measures together allows the analysis to determine whether a candidate that appears attractive under average preferences also remains defensible under less favourable decision conditions.

Evidence 16 - Decision-Rule Comparison

	Decision_Rule	Variant	Mean_Regret	CVaR95_Regret	Worst_Case_Regret	Zero_Regret_Frequency
0	Existing_TOPSIS_reference	XGBoost class_weight isotonic	0.033974	0.455535	0.933845	0.8576
1	Minimax_regret	XGBoost class_weight platt	0.096088	0.446101	0.887321	0.0000
2	CVaR95_regret	XGBoost class_weight platt	0.096088	0.446101	0.887321	0.0000
3	Minimum_mean_regret	XGBoost class_weight isotonic	0.033974	0.455535	0.933845	0.8576

Figure 16. Comparison of deployment recommendations produced by alternative decision rules.

This experiment compares the recommendations generated by the previous preference-based approach with the subsequent regret-aware decision rules.

The purpose is not to claim that one decision rule is universally superior. Instead, the comparison examines whether the final recommendation is sensitive to the assumptions embedded in the decision rule.

If different rules select different candidates, this provides evidence that deployment selection is affected by decision-policy uncertainty and that reporting only one weighted recommendation could conceal an important source of uncertainty.

Evidence 17 - Bootstrap Selection Stability

	Decision_Rule	Variant	Selection_Frequency
1	CVaR95_Winner	XGBoost class_weight platt	1.0
2	Mean_Regret_Winner	XGBoost class_weight isotonic	1.0
0	Minimax_Winner	XGBoost class_weight platt	1.0

Figure 17. Bootstrap analysis of candidate-selection stability under repeated resampling.

The bootstrap analysis evaluates whether the recommended candidate is repeatedly selected when the available evidence is perturbed through resampling.

This provides a complementary form of validation.

A candidate that achieves favourable regret but is selected only rarely across bootstrap repetitions would provide weaker evidence for a reproducible recommendation than a candidate that remains frequently selected.

The bootstrap analysis therefore distinguishes decision quality from decision reproducibility.

7.5 Joint Preference and Corruption Uncertainty

An additional experiment extends the regret analysis by considering preference uncertainty and sensor-corruption uncertainty jointly.

	Variant	Mean_Joint_Regret	P95_Joint_Regret	CVaR95_Joint_Regret	Worst_Case_Joint_Regret	Zero_Regret_Frequency	Minimax_Rank	CVaR95_Rank	Mean_Regret_Rank
3	XGBoost class_weight platt	0.123966	0.560538	0.695729	0.969285	0.089765	1	2	2
2	XGBoost class_weight isotonic	0.084179	0.534891	0.684579	0.971801	0.686618	2	1	1
0	Random Forest class_weight isotonic	0.533107	0.981915	0.987313	0.998236	0.043500	3	3	4
1	Random Forest class_weight platt	0.530673	1.000000	1.000000	1.000000	0.180118	4	4	3

Figure 18. Tail-regret analysis under joint stakeholder-preference and sensor-corruption uncertainty.

This experiment is important because preference uncertainty and data-quality uncertainty do not necessarily occur independently in a real deployment setting.

A candidate may appear attractive under clean sensor conditions but become less desirable when sensor quality deteriorates. Similarly, the consequences of that deterioration may depend on which deployment objective the stakeholder considers most important.

The joint analysis therefore asks:

Which candidate remains comparatively defensible when both the operating evidence and the stakeholder preference are uncertain?

This creates a direct connection between the sensor-corruption evidence from Section 5 and the downstream decision analysis.

7.6 Minimax-Regret Analysis

The minimax-regret analysis provides a conservative decision rule for situations in which the true deployment preferences or operating conditions cannot be known with confidence.

Rather than selecting the configuration that performs best under one fixed preference setting, minimax regret evaluates the opportunity loss associated with each candidate across the considered scenarios. The preferred candidate is the one with the smallest worst-case regret, thereby limiting the maximum loss that could result from selecting a configuration under uncertainty.

The output below analyses examine decision risk under preference uncertainty, joint preference-and-corruption uncertainty, and pipeline-level deployment alternatives.

Evidence 19 - AI4I Regret Analysis under Preference Uncertainty

	Variant	Mean_Regret	P95_Regret	Worst_Case_Regret	Zero_Regret_Frequency	Robust_Regret_Rank
3	XGBoost class_weight platt	0.096088	0.287290	0.887321	0.0000	1
2	XGBoost class_weight isotonic	0.033974	0.278828	0.933845	0.8576	2
0	Random Forest class_weight isotonic	0.447315	0.873045	0.991296	0.0740	3
1	Random Forest class_weight platt	0.538289	0.969837	0.999690	0.0684	4

RMRS recommendation: XGBoost | class_weight | platt

Figure 19. Regret-risk comparison of leading AI4I deployment configurations under uncertain deployment preferences.

Among the four shortlisted AI4I configurations, XGBoost | class_weight | Platt achieved the lowest worst-case regret of 0.8873, compared with 0.9338 for XGBoost with isotonic calibration, 0.9913 for Random Forest with isotonic calibration, and 0.9997 for Random Forest with Platt calibration.

Accordingly, the robust-regret analysis ranks XGBoost | class_weight | Platt first.

The result also shows why multiple regret measures are useful. XGBoost | class_weight | isotonic has the lowest mean regret at approximately 0.0340, compared with approximately 0.0961 for the Platt variant. However, the Platt variant has the lower worst-case regret.

This reveals a genuine decision trade-off: the configuration with the best average outcome is not necessarily the configuration that provides the strongest protection against the worst-case preference scenario.

Evidence 20 - AI4I Joint Preference and Corruption Uncertainty

	Variant	Mean_Joint_Regret	P95_Joint_Regret	Worst_Case_Joint_Regret	Zero_Regret_Frequency	Joint_Robust_Regret_Rank
3	XGBoost class_weight platt	0.123966	0.560538	0.969285	0.089765	1
2	XGBoost class_weight isotonic	0.084179	0.534891	0.971801	0.686618	2
0	Random Forest class_weight isotonic	0.533107	0.981915	0.998236	0.043500	3
1	Random Forest class_weight platt	0.530673	1.000000	1.000000	0.180118	4

Preference + corruption robust recommendation: XGBoost | class_weight | platt

Figure 20. Joint regret analysis of AI4I configurations under simultaneous preference and sensor-corruption uncertainty.

When preference uncertainty and sensor-corruption uncertainty are considered jointly, XGBoost | class_weight | Platt remains the first-ranked robust-regret configuration, with a worst-case joint regret of approximately 0.9693.

The XGBoost isotonic alternative achieves a slightly larger worst-case joint regret of approximately 0.9718, while the Random Forest configurations show higher values of approximately 0.9982-1.0000.

Therefore, the explicit recommendation generated by this analysis is:

Preference + corruption robust recommendation: XGBoost | class_weight | Platt

However, the results again demonstrate an important distinction between average and conservative decision criteria. XGBoost | class_weight | isotonic achieves the lower mean joint regret (0.0842) and lower P95 joint regret (0.5349) than the Platt alternative (0.1240 mean and 0.5605 P95), while the Platt configuration achieves the slightly lower worst-case regret.

Thus, the evidence does not suggest that all risk measures are dominated by a single configuration. Instead, it demonstrates that the choice of risk attitude determines the ultimate suggestion.

Evidence 21 - Pipeline-Level Minimax-Regret Analysis

	Pipeline	Mean_Regret	P95_Regret	Worst_Case_Regret	Zero_Regret_Frequency	Robust_Regret_Rank
1	Isolation Forest gate	0.127121	0.442141	0.787279	0.3616	1
3	Isolation Forest gate + periodic 10	0.133400	0.426978	0.788021	0.0000	2
2	Isolation Forest gate + periodic 5	0.142006	0.420082	0.789159	0.0000	3
4	Isolation Forest gate + periodic 20	0.148313	0.441919	0.789580	0.0080	4
0	Always-on RUL inference	0.130659	0.513744	0.896898	0.4556	5
6	LSTM Autoencoder gate + periodic 5	0.323502	0.723760	0.959882	0.0652	6
7	LSTM Autoencoder gate + periodic 10	0.325757	0.741985	0.964285	0.0166	7
5	LSTM Autoencoder gate	0.324007	0.748419	0.968577	0.0930	8
8	LSTM Autoencoder gate + periodic 20	0.374820	0.811496	0.988361	0.0000	9

Figure 21. Regret-risk comparison of C-MAPSS deployment-pipeline alternatives under uncertain decision conditions.

So, the robust-regret criterion ranks the Isolation Forest gate first at the pipeline level because it delivers the lowest worst-case regret of around 0.7873.

The closely related periodic Isolation Forest alternatives also perform competitively, with worst-case regret values of approximately 0.7880-0.7896.

In comparison, Always-on RUL inference produces a higher worst-case regret of approximately 0.8969, while the LSTM Autoencoder gate alternatives produce substantially higher worst-case regret values of approximately 0.9599-0.9884.

These results indicate that the regret framework can also distinguish between alternative deployment pipelines rather than being restricted to individual predictive-model configurations.

Crucially, the outcome should be seen as deployment advice that varies depending on the circumstance. It does not prove that the Isolation Forest gate is always the optimal pipeline. Instead, it restricts worst-case decision loss more successfully than the assessed alternatives under the experiment's objectives, assumptions, and uncertainty situations.

7.6.1 Overall Interpretation of the Minimax-Regret Evidence

Therefore, Figures 19-21 demonstrate that deployment selection can change depending on how decision risk is defined.

In the AI4I analysis, XGBoost | class_weight | isotonic provides the lowest average regret, whereas XGBoost | class_weight | Platt provides the lowest worst-case regret. When preference and corruption uncertainty are considered jointly, the Platt configuration remains the minimax-regret recommendation, although the difference in worst-case regret between the two XGBoost alternatives is small.

At the pipeline level, the Isolation Forest gate provides the lowest worst-case regret among the evaluated alternatives.

These findings shows why the downstream analysis should not simply identify the candidate with the highest predictive score or the best result under one fixed preference vector. Different configurations may be preferable under average-risk, tail-risk and worst-case-risk perspectives.

The minimax-regret analysis therefore provides a conservative downstream decision rule:

Among the defensible candidates, select the configuration that minimises the largest opportunity loss across the evaluated uncertainty scenarios.

This supports a risk-aware deployment recommendation while preserving an important limitation: the resulting winner is conditional on the objectives, candidate configurations, preference scenarios, corruption conditions and operational assumptions included in the study.

7.7 Impediment Identified

The main methodological issue identified at this stage concerns the interpretation of regret-based recommendations.

A minimax-regret winner should not automatically be described as the universally best predictive-maintenance model. Its selection depends on the candidate set, deployment objectives, preference scenarios, corruption scenarios and operational assumptions included in the analysis.

Besides, assumed operational penalties are software-level decision assumptions rather than measured real-world maintenance costs. They then need to be clearly identified as scenario assumptions when interpreting the results.

8. Supervisor Feedback and Main Research Impediment

During supervision, the main issue identified was methodological positioning rather than lack of technical work.

The supervisor questioned:

- why a consensus ranking is needed
- how RDRA should be justified relative to Borda
- what property makes one ranking preferable
- whether ranking stability is always desirable
- how predictive robustness differs from ranking robustness
- whether ranking and robustness can be connected
- whether the project needs several separate novelty claims

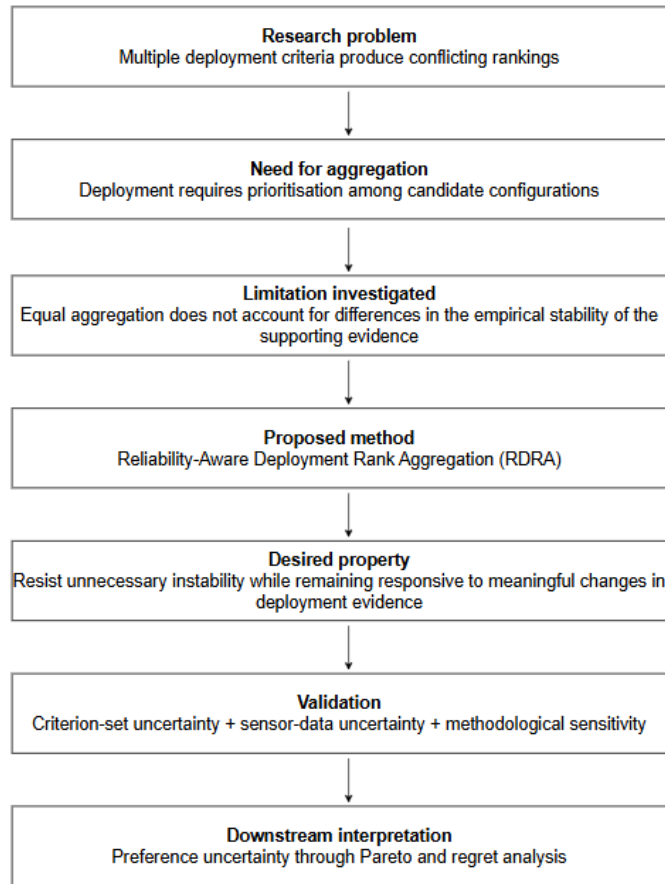
The supervision process made clear that experimental quantity alone does not establish a research contribution. The project required a clearer explanation of why the proposed method should exist and what property it is intended to improve.

9. Corrective Action Following Supervisor Feedback

Following the supervision meeting, I refined the following elements of the research:

- the working research title
- the central research problem
- the justification for consensus ranking
- the role of Equal Borda as a baseline
- the specific limitation investigated by RDRA
- the definition of desirable ranking behaviour
- the research objective
- the main research question
- the distinction between predictive robustness and ranking robustness
- the role of sensor corruption as a validation mechanism
- and the role of Pareto/regret analysis as downstream decision analysis rather than a separate central novelty.

Following the supervision discussion, the project was reorganised around one central methodological contribution.



10. Revised Research Direction

Working Title

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

Research Objective

To develop and empirically validate a reliability-aware rank-aggregation methodology for deployment-oriented predictive-maintenance model selection that accounts for differences in the stability of evaluation evidence and produces a defensible consensus ranking under evaluation uncertainty.

Main Research Question

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable evaluation evidence?

Supporting Validation Perspectives

Criterion / evidence uncertainty

How does RDRA compare with Equal Borda and the existing composite method when the deployment-criterion set changes?

Data-quality uncertainty

How does the consensus ranking behave when deployment evidence is changed through controlled sensor-data corruption?

Preference uncertainty

How does uncertain stakeholder preference affect downstream selection among candidates supported by the validated evidence?

11. Current Contribution and Validation Logic

The current primary research contribution is:

A reliability-aware deployment rank-aggregation methodology for predictive-maintenance model selection that incorporates empirical stability of heterogeneous evaluation evidence and evaluates the resulting consensus under criterion-set and sensor-data uncertainty.

The contribution is not claimed to be a new predictive algorithm; the invention of Borda, a new corruption technique, a new Pareto method or a new regret algorithm.

Instead, the contribution concerns the adaptation, integration and empirical validation of reliability-aware rank aggregation in deployment-oriented predictive-maintenance selection. Your existing report already makes this boundary explicit.

The desired properties are:

- Consistency - avoid disproportionate ranking changes under reasonable evidence variation.
- Reliability awareness - recognise differences in empirical stability of criterion-level evidence.
- Responsiveness - remain capable of changing when meaningful deployment evidence changes.

The method is not assumed to be superior. If Equal Borda performs equally well or better under the relevant conditions, the additional RDRA complexity would not be justified.

12. Current Status and Next Steps

At the end of Check Point 2, the project has completed or substantially completed:

- AI4I classification experiments
- C-MAPSS RUL experiments
- deployment-oriented evaluation
- sensor-corruption analysis
- RDRA implementation
- Equal Borda comparison
- criterion-subset perturbation
- leave-one-dimension-out sensitivity
- deployment-rank stability analysis
- normalisation sensitivity
- Pareto and regret analysis
- bootstrap and joint uncertainty analysis
- methodological refinement following supervisor feedback

Therefore, the project has progressed from model benchmarking to methodological development and to validation and consolidation.

The next priorities are:

1. Formalise RDRA mathematically, including the exact stability-weight calculation.
2. Strengthen literature positioning against weighted, robust, reliability-aware and stability-based rank aggregation.
3. Map every experiment to a specific research claim.

13. Checkpoint Reflection and Summary

This checkpoint produced an important shift in the project.

At first, experimental was the main indicator of progress. However, weekly supervision made clear that many findings do not prove research uniqueness on their own. Finding a well-founded methodological issue, outlining the necessity of the suggested approach, defining its intended characteristics, and conducting tests to verify those characteristics are the most crucial requirements.

Therefore, the main learning from this checkpoint was to establish why the proposed method should exist before using experimental results to argue that it works.

The project is now organised around one coherent contribution rather than several independent novelty claims.

The checkpoint demonstrates the required progress in three areas:

- Research progress - substantial experimental and methodological milestones have been achieved.
- Evidence of achievements - progress is supported by notebook outputs, ranking comparisons, robustness results, sensitivity analyses and decision-risk evidence.
- Impediment identification and supervisor involvement - methodological concerns were identified through weekly supervision and used to refine the problem, objective, RQ, method, validation logic and contribution.

References

- [1] T. P. Carvalho, F. A. A. M. N. Soares, R. Vita, R. da P. Francisco, J. P. Basto, and S. G. S. Alcalá, "A systematic literature review of machine learning methods applied to predictive maintenance," *Computers & Industrial Engineering*, vol. 137, Art. no. 106024, 2019.
- [2] A. Theissler, J. Pérez-Velázquez, M. Kettelgerdes, and G. Elger, "Predictive maintenance enabled by machine learning: Use cases and challenges in the automotive industry," *Reliability Engineering & System Safety*, vol. 215, Art. no. 107864, 2021.
- [3] O. Serradilla, E. Zugasti, J. Rodriguez, and U. Zurutuza, "Deep learning models for predictive maintenance: A survey, comparison, challenges and prospects," *Applied Intelligence*, vol. 52, no. 10, pp. 10934–10964, 2022.
- [4] X. Bouthillier et al., "Accounting for variance in machine learning benchmarks," *Proceedings of Machine Learning and Systems*, vol. 3, 2021.
- [5] UCI Machine Learning Repository, "AI4I 2020 Predictive Maintenance Dataset," 2020.
- [6] A. Fernández, S. García, F. Herrera, and N. V. Chawla, "SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary," *Journal of Artificial Intelligence Research*, vol. 61, pp. 863–905, 2018.
- [7] M. Kull, T. M. Silva Filho, and P. Flach, "Beyond sigmoids: How to obtain well-calibrated probabilities from binary classifiers with beta calibration," *Electronic Journal of Statistics*, vol. 13, no. 2, pp. 5053–5080, 2019.
- [8] A. Saxena and K. Goebel, "Turbofan Engine Degradation Simulation Data Set," NASA Ames Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2008.
- [9] B. Taşçı, A. Omar, and S. Ayvaz, "Remaining useful lifetime prediction for predictive maintenance in manufacturing," *Computers & Industrial Engineering*, vol. 184, Art. no. 109566, 2023.

- [10] A. Prasad, H. Pareek, and P. Ravikumar, "Distributional rank aggregation, and an axiomatic analysis," in Proceedings of the 32nd International Conference on Machine Learning, vol. 37, pp. 2104–2112, 2015.
- [11] V. Pihur, S. Datta, and S. Datta, "RankAggreg, an R package for weighted rank aggregation," BMC Bioinformatics, vol. 10, Art. no. 62, 2009.
- [12] M. G. Kendall, "A new measure of rank correlation," Biometrika, vol. 30, nos. 1–2, pp. 81–93, 1938.
- [13] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in Proc. International Conference on Learning Representations (ICLR), 2019.
- [14] S. K. Singh and R. Kumar, "A hybrid multi-criteria decision-making model for Industry 4.0 technology evaluation using Pareto and TOPSIS analysis," SN Applied Sciences, vol. 6, no. 3, 2024.